OrbitReady: A Source-Linked Pre-Review Copilot for Spacecraft Debris-Mitigation Evidence

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Team TwinForge · Hackyard Build 2026 · IIT Guwahati · 24–25 October 2026

Corrected and scope-controlled edition · 17 October 2026

Safety boundary: OrbitScore is not an official certification tool, regulator, collision-prediction system, operator-facing conjunction-alert service, or replacement for NASA Debris Assessment Software. Its readiness band is a prioritization index, not a scientific safety probability.

Abstract

Small-spacecraft teams must assemble mission parameters, disposal evidence, passivation evidence, and jurisdiction-specific guidance before expert or regulatory review. These facts are often distributed across documents and spreadsheets. We present OrbitScore, a prototype source-linked pre-review copilot that accepts or extracts mission facts, applies versioned deterministic rule fixtures, identifies missing evidence, compares a controlled counterfactual, and exports a cited review packet. The prototype separates document assistance from rule evaluation and requires human review for critical findings. Demonstrations use labeled synthetic mission profiles and public guidance. OrbitScore is not a regulator, certification tool, collision-prediction system, operator-facing conjunction-alert service, or replacement for NASA Debris Assessment Software. The prototype's readiness band is a transparent prioritization index, not a validated safety probability.

Keywords: orbital debris; post-mission disposal; evidence traceability; human-in-the-loop; counterfactual explanation; provenance; mission review

1. Introduction

Orbital debris mitigation is a real engineering and sustainability problem. Foundational work by Kessler and Cour-Palais and later environment studies examined how collisions and fragmentation can increase the long-term debris population and affect future operations [1–3]. These studies support mitigation as a design concern, but they do not supply a universal readiness score for every spacecraft or jurisdiction.

The practical problem addressed here is narrower: a small spacecraft team may have a mission concept, orbit, propulsion plan, and several technical documents, but no single trace connecting each claim to its source, scope, date, and reviewer action. A late discovery that a disposal method is unproven or that a document conflicts with a spreadsheet can create rework. OrbitReady is designed as an evidence-readiness layer before specialist analysis, filing, or operational services. OrbitScore names the product's internal prioritization heuristic.

The central product thesis is therefore traceability rather than prediction. The system should answer: What is known? What is only asserted? What evidence is missing? Which source profile applies? What controlled design change would alter the review?

2. Problem and Design Objective

2.1 The operator bottleneck

The relevant workflow crosses mission design, propulsion, spacecraft configuration, disposal planning, documentation, and policy interpretation. NASA's orbital-debris standard describes separate technical areas for normal releases, explosions and breakups, collisions, postmission disposal, and reentry casualty risk [4]. Public FCC, NASA, ESA, ISO, UN, IADC, and ISRO materials do not form one interchangeable rulebook. A review tool must preserve the difference between a regulation, an agency standard, an international guideline, a public initiative, and an internal assumption.

OrbitScore is intended for a systems engineer, mission designer, or university project team preparing an internal packet. It is not intended to make an operational maneuver decision, approve a mission, or replace an agency assessment.

2.2 Design objective

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mission facts → evidence and provenance → scoped profile → deterministic findings
→ missing proof / reviewer action → controlled counterfactual → cited pre-review packet
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3. Scope, Policy Profiles, and Safety Boundaries

OrbitScore stores policy profiles as versioned fixtures. A profile contains the organization, jurisdiction or scope, document version, clause reference, input fields, rule expression, evidence requirement, and human-review note. The system must never silently merge rules from different profiles.

Profile/source	What it supports	What OrbitScore may do	What it must not claim
FCC 22-74	U.S. FCC disposal rule for covered space stations in defined LEO/uncontrolled-reentry scope	Show a clearly labeled FCC-inspired evidence check	Universal global five-year law or regulator approval
NASA-STD-8719.14C / DAS context	NASA program/project debris-assessment requirements and methods	Link evidence fields to a NASA-inspired pre-review fixture	NASA compliance, NASA endorsement, or DAS replacement
UN COPUOS / IADC	International guidance and technical mitigation principles	Store source-linked guidance and reviewer questions	Binding law in every jurisdiction
ISO 24113:2023	International standard scope and requirements where accessible and applicable	Record version/scope and verified clauses	Full conformance without access to and interpretation of the standard
ESA ESSB-ST-U-007	ESA-specific debris-mitigation requirements	Use an ESA-scoped profile when exact clauses are available	Universal European or global regulation
ISRO DFSM statement	Public Indian intent for debris-free missions by 2030 and described practices	Provide an India DFSM-inspired checklist with source labels	ISRO endorsement or a universal Indian certificate

The FCC order states that the five-year requirement applies to covered FCC space stations ending their mission in or passing through the LEO region below 2,000 km and planning disposal through uncontrolled atmospheric re-entry, within the FCC's licensing and market-access framework. OrbitReady therefore presents this only as a configured FCC 22-74 profile for the specified scope; other profiles may use different

disposal criteria. It defines end of mission and includes transition provisions [5]. This is materially narrower than saying all satellites everywhere must deorbit within five years.

NASA-STD-8719.14C is a NASA standard that provides technical requirements and methods for limiting debris. Requirement 4.3-1 addresses planned debris released during deployment, operation, or disposal while passing through LEO: debris at least 1 mm must meet the 25-year lifetime condition and the total object-time product must be below 100 object-years per launch vehicle upper stage or spacecraft. Other separate requirements address accidental collisions, postmission disposal, and reentry risk [4]. OrbitReady stores these as NASA-scoped interpretation fixtures, not as a public universal compliance API.

ISRO's official DFSM page describes an initiative aiming for debris-free space missions by Indian space actors by 2030. It describes more than 99% postmission-disposal success and less than five years of remaining orbital life for specified rocket bodies and spacecraft at end of operational life, along with trackability, identifiability, maneuverability, coordination, and data sharing [6]. OrbitScore treats this as a scoped public-source profile, not as an endorsement or universal statutory rule.

4. System Architecture

The architecture separates deterministic evaluation from document assistance. The language model may help locate and summarize evidence, but it does not calculate orbital decay, collision probability, casualty risk, or a rule boundary.

4.1 Layer 1 — intake and evidence capture

- Accept structured mission inputs and document uploads. Every value stores source type, title, URL or file identifier, page/section where available, source date, retrieval date, unit, and extraction confidence.
- Use layout-aware extraction only as an assistive path. A human can correct extracted values; corrections are preserved as edits rather than silently replacing the source.
- Use public GP/TLE/OMM records only for contextual or historical display. Store epoch and age. Do not make live orbital feeds a dependency for the core review.

4.2 Layer 2 — deterministic profile evaluator

- Normalize units and validate required fields.
- Apply a versioned profile rule and return Pass under profile, Warning, Needs evidence, Not evaluated, or Conflict.
- Block a Pass when a critical input is missing, contradictory, stale beyond the configured limit, or outside profile scope.

4.3 Layer 3 — explanation, counterfactual, and packet

- Generate a source-linked explanation from stored findings and clauses. Retrieved text constrains the explanation but does not guarantee truth.
- Run controlled what-if comparisons over explicitly permitted parameters. Counterfactuals are illustrative design trades, not causal guarantees or flight recommendations.
- Export a packet containing assumptions, findings, evidence links, unresolved gaps, reviewer actions, profile version, and limitations.

5. Evidence States and Readiness Band

A single weighted score can hide a critical unknown. The prototype therefore uses hard evidence states first and a secondary band only for prioritization. The band is not a scientific probability, risk estimate, certification status, or regulatory result.

State	Meaning	Example reviewer action
Pass under profile	Required input and evidence are present and the configured rule evaluates positively	Confirm source version and retain evidence
Warning	The rule evaluates but a limitation, age, or scope issue remains	Review assumptions and update evidence
Needs evidence	A required claim is missing proof or provenance	Assign evidence owner and due date
Conflict	Two sources or inputs disagree	Resolve with qualified reviewer; do not average
Not evaluated	Profile or data is outside supported scope	Escalate to expert or select a valid profile

If a band is shown, its inputs should be evidence completeness and unresolved critical findings, not an invented physical probability. Suggested labels are Ready for expert review, Review gaps first, and Not evaluable. The band must never override a critical Needs evidence or Not evaluated state.

6. Illustrative Synthetic Counterfactual Design Review

Counterfactual explanations are useful when a reviewer asks what change would alter a decision [7]. OrbitReady implements a constrained comparison over a small set of declared variables, such as disposal lifetime assumption, disposal evidence present/absent, or a documented end-of-life maneuver capability. The engine returns the changed input, affected rule, before/after state, and assumptions. This is an illustrative synthetic analysis of the configured rule fixtures, not a flight recommendation or validated physical prediction.

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before: disposal_lifetime_years = 8.4; evidence = missing
after:  disposal_lifetime_years = 4.2; evidence = synthetic analysis attached
result:  Needs evidence → Pass under configured profile
warning: illustrative fixture; not a flight recommendation
```

A physically meaningful trade requires ballistic coefficient or area-to-mass ratio, mass properties, atmospheric model, solar and geomagnetic assumptions, epoch, orbit geometry, propulsion performance, execution reliability, and disposal-success evidence. If those inputs are not available, OrbitReady presents a qualitative or evidence-only counterfactual, not a precise burn recommendation. The synthetic example below illustrates how configured assumptions affect the internal comparison; it does not provide an explicit trade envelope for flight operations.

SHAP and LIME are not required for a deterministic rule trace. For threshold-driven findings, an exact trace is clearer and less misleading: input value, threshold, source clause, status, and reviewer action. Attribution methods may be added later for a learned extraction or ranking model, but they must not be presented as causal engineering explanations.

7. Provenance, Confidence, and Human Review

Provenance is a system property. A provenance record should answer why a finding was produced and where each input came from. W3C PROV-O and database provenance literature provide useful conceptual

models [8]. SHA-256 hashes can make changes tamper-evident when a controlled storage and anchoring process exists; a hash alone does not make a database immutable or legally authoritative.

The prototype uses evidence confidence rather than pretending to have a calibrated probability. High confidence means an authoritative, current source, complete required fields, and no conflict. Medium confidence means a source exists but is stale, incomplete, or secondary. Low confidence means an assumption, extraction uncertainty, conflict, or unsupported scope. Model calibration research shows that raw confidence can be overconfident, especially under shift [9,10].

Human review is mandatory for critical findings, conflicts, profile selection, and any recommendation that could alter mission design. The system recommends; a qualified human decides. It does not file, approve, command, or fire a thruster. The following is an internal OrbitReady automation-boundary mapping inspired by Parasuraman et al. [16], not a certification against that taxonomy.

Automated assistance	Human-led decision
Parse candidate values from documents	Confirm the value and its meaning
Normalize units and run deterministic checks	Select the applicable jurisdiction/profile
Show evidence gaps and controlled what-if results	Accept engineering assumptions and tradeoffs
Assemble a cited packet	Approve any submission or operational action

8. Implementation Status

Capability	Status in corrected paper/demo	Boundary
Structured mission intake and seeded synthetic fixtures	Implemented in the described MVP workflow	Synthetic/demo data; not operational mission evidence
Versioned deterministic rule fixtures and evidence states	Implemented as the core design and test target	Profile-scoped; not official compliance
Source-linked provenance and cited packet structure	Prototype	Hashing and links support traceability but do not create legal immutability
Document parsing and optional grounded extraction	Prototype	Extraction remains candidate data until human approval
OrbitScore readiness prioritization heuristic	Prototype	Evidence-prioritization index; not a safety probability
Operational conjunction assessment, maneuver advice, NASA DAS/ESA DRAMA execution	Planned / not implemented	Explicitly outside MVP and not represented as current capability
King-Hele, Jacchia-Roberts, NRLMSISE-00 atmospheric propagation	Planned / not implemented	No physical decay claim is made by the corrected MVP
Regulatory filing, certification, or regulator integration	Planned / not implemented	Requires qualified review, applicable scope, and authorized tooling

The status labels above separate what the paper demonstrates from what remains a roadmap. “Prototype” means a bounded proof-of-concept workflow, not production validation. “Planned / not implemented” items must not be described as current system outputs.

9. Data Sources and Restrictions

Source	Useful fields/content	Use in prototype	Restriction / limitation
User mission profile	Orbit, altitude, mass, duration, propulsion, disposal, evidence links	Primary input	May be incomplete, private, or contradictory
NASA DAS / ORDEM	Assessment context; debris flux and model outputs	Reference or future integration	Specialized access, versioned models, not a live CA feed
ESA DISCOSweb	Launch, object registration, spacecraft	Optional reference lookup	Terms/account; ephemeris is not distributed via DISCOSweb

	size/mass/shape/mission/owner		[11]
Space-Track GP/OMM	Catalog identifiers, epoch, mean elements, OMM fields	Historical/contextual only	Account/terms; public TLE is not sufficient for operational CA
CelesTrak GP	CATNR, INTDES, GROUP, NAME; TLE/OMM XML/KVN/JSON/CSV	Educational/contextual only	Rate limits, age, format limitations; cache and stop on errors [12]
Official policy documents	Clauses, scope, version, dates	Versioned profile fixtures	Guidelines and standards are not interchangeable
Synthetic mission fixtures	Complete, missing, conflict, stale, failure cases	Required for reproducible demo	Must be labeled synthetic

10. Validation Plan

The 24-hour validation target is reproducibility of the workflow, not scientific validation of a safety model. A versioned JSON baseline should contain inputs, expected states, source references, assumptions, and reviewer actions.

Test	Pass condition
Complete fixture	Expected findings and citations match baseline
Missing disposal evidence	Returns Needs evidence or Not evaluated; never Pass
Boundary value	Behavior at configured threshold is explicit and stable
Conflicting documents	Conflict is shown; no silent merge
Stale external record	Age is visible and status is downgraded or blocked
API timeout / 403 / 404	Core review continues with labeled cache or Not evaluated; no retry storm
Unsupported profile	Not evaluated plus expert escalation
Unsupported explanation	Generated text is rejected or visibly marked unsupported
Counterfactual	Only declared inputs and affected findings change
Determinism	Same input and rule version produce same output
Export integrity	Every finding has input and source references

The prototype cannot establish regulatory acceptance, operational conjunction-assessment performance, generalization across jurisdictions, real mission safety, or time savings across organizations. Those are future validation questions.

11. Synthetic Demonstrations

The demonstrations below are intentionally labeled synthetic. They illustrate workflow behavior and are not evidence about a real mission.

11.1 Complete passive-deorbit fixture

A synthetic 3U profile contains a source-linked orbit, documented area-to-mass assumption, a disposal-lifetime analysis, and a passivation statement. OrbitScore returns Pass under the selected internal profile for the represented checks, with a note that expert review remains required. It does not output APPROVED.

11.2 Missing-evidence fixture

A synthetic 12U profile includes a disposal lifetime assertion but no analysis source and a conflicting propulsion value in a second document. OrbitScore returns Needs evidence and Conflict. The packet identifies the two sources, prevents averaging, and assigns a reviewer action.

11.3 Controlled counterfactual fixture

The user attaches a synthetic disposal analysis under an explicitly configured assumption. The before/after comparison changes only the disposal finding and evidence band. It does not recommend a burn, predict a conjunction, or claim that the mission is safe.

12. Related Work and Positioning

NASA DAS and ORDEM, ESA DRAMA and DISCOS, Space-Track services, and commercial orbital-intelligence products address specialized assessment, environment modeling, reference data, or operational space-domain awareness. OrbitScore should not compete with those systems on sensors, catalog authority, or conjunction operations. Its proposed gap is the earlier evidence handoff: source-linked mission facts, scoped rule fixtures, missing-proof detection, controlled what-if comparison, and a cited internal review packet.

Commercial product pages are used here only as product references. They are not scientific evidence for debris growth, regulatory scope, or the validity of the OrbitScore band.

13. Limitations and Ethical Boundaries

- OrbitScore is not an official certification or compliance authority.
- OrbitScore is not a collision-prediction or operator-facing conjunction-alert system.
- Public orbital data is contextual only and must carry an epoch, age, source, and usage label.
- Synthetic data is not mission evidence.
- A readiness band is not a safety probability or legal status.
- Counterfactuals are controlled comparisons, not causal guarantees or flight commands.
- The product cannot resolve conflicting standards without a qualified human reviewer.

14. Hackathon Implementation Blueprint

Build area	24-hour deliverable	Deferred work
Intake	Structured form with provenance and synthetic PDF	Secure enterprise connectors
Rules	6–10 deterministic checks across two or three clearly labeled profiles	Full jurisdiction library
Review	Pass / Warning / Needs evidence / Conflict / Not evaluated	Operational approval workflow
Counterfactual	One controlled before/after variable	Multi-variable optimization
Packet	Cited Markdown/HTML export	Regulatory filing formats
AI	Optional grounded extraction and explanation	Model training and calibration study
Data	Synthetic fixtures plus official public documents	Live orbital-feed dependency

The strongest demo moment is simple: a mission begins with a missing disposal proof. OrbitScore refuses to pass it, shows the exact source-linked gap, lets the user attach evidence or change the scoped assumption, recomputes the finding, and exports the trace. Judges can inspect the input, rule, source, limitation, and reviewer action in one flow.

15. Conclusion

The defensible contribution is not an autonomous compliance engine or a new collision-probability model. It is a source-linked evidence-readiness workflow that keeps deterministic rules, document assistance, uncertainty, provenance, and human review separate. The corrected prototype can be technically credible if it uses scoped profiles, labeled synthetic fixtures, conservative statuses, reproducible tests, and explicit limitations.

Future work may integrate formal agency tools or authorized operational services, but only after data rights, model validity, mission-specific engineering review, and the applicable regulatory context are established. Until then, OrbitScore should remain a pre-review assistant that improves the quality of expert handoff rather than replacing it.

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